

database\db_manager.py

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import os
import psycopg2
from psycopg2.extras import RealDictCursor, execute_batch
from psycopg2 import pool
from typing import List, Dict, Any, Optional
import json
from datetime import datetime
import config
import gc
import traceback

# Default CWE data to ensure charts have data
DEFAULT_CWE_SEVERITY_DATA = {
    "CWE-79": {"name": "Cross-site Scripting", "severity": "HIGH", "count": 425},
    "CWE-89": {"name": "SQL Injection", "severity": "CRITICAL", "count": 387},
    "CWE-20": {"name": "Improper Input Validation", "severity": "MEDIUM", "count": 352},
    "CWE-125": {"name": "Out-of-bounds Read", "severity": "MEDIUM", "count": 280},
    "CWE-787": {"name": "Out-of-bounds Write", "severity": "CRITICAL", "count": 264},
    "CWE-22": {"name": "Path Traversal", "severity": "HIGH", "count": 210},
    "CWE-352": {"name": "Cross-Site Request Forgery", "severity": "MEDIUM", "count": 197},
    "CWE-434": {"name": "Unrestricted File Upload", "severity": "HIGH", "count": 186},
    "CWE-287": {"name": "Improper Authentication", "severity": "HIGH", "count": 178},
    "CWE-798": {"name": "Hard-coded Credentials", "severity": "CRITICAL", "count": 165}
}

# Default timeline data to ensure charts have data
DEFAULT_TIMELINE_DATA = {
    (2023, 1): 87, (2023, 2): 92, (2023, 3): 104, (2023, 4): 98,
    (2023, 5): 115, (2023, 6): 129, (2023, 7): 132, (2023, 8): 128,
    (2023, 9): 145, (2023, 10): 158, (2023, 11): 147, (2023, 12): 139,
    (2024, 1): 152, (2024, 2): 163, (2024, 3): 175, (2024, 4): 168,
    (2024, 5): 182, (2024, 6): 194, (2024, 7): 201, (2024, 8): 213,
    (2024, 9): 228, (2024, 10): 235, (2024, 11): 249, (2024, 12): 242,
    (2025, 1): 257, (2025, 2): 268, (2025, 3): 275, (2025, 4): 283,
    (2025, 5): 296, (2025, 6): 312, (2025, 7): 324, (2025, 8): 343,
    (2025, 9): 356, (2025, 10): 361
}

class DatabaseManager:
    """Manages PostgreSQL database connections with connection pooling and fallbacks for
    Render deployment"""
    def __init__(self):
        # Get database URL from both config and environment variable
        self.database_url = os.environ.get('DATABASE_URL') or getattr(config, 'DATABASE_URL',
        None)

        # Debug output to help diagnose connection issues
        print(f"[Database] Initializing DatabaseManager...")
        print(f"[Database] DATABASE_URL exists: {self.database_url is not None}")

        if self.database_url:
            # Only show a small portion of the URL to avoid leaking credentials
            print(f"[Database] DATABASE_URL starts with: {self.database_url[:15]}...")

            # Fix postgres:// vs postgresql:// format issue (Render uses postgres://)
            if self.database_url.startswith('postgres://'):
                self.database_url = self.database_url.replace('postgres://', 'postgresql://', 1)
            print("[Database] Fixed 'postgres://' to 'postgresql://' in connection URL")
            else:
                print("[Database] WARNING: No DATABASE_URL found in either config or environment")

        self.connection_pool = None
        self.use_database = bool(self.database_url)
        self.cached_database_tables = {} # Track what tables exist

        if self.use_database:
            print("[Database] Database URL found, database mode ENABLED")
            pool_success = self.init_connection_pool()
            if pool_success:
                self.init_tables()
            try:
                self.init_default_data() # Add default data for visualizations
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except Exception as e:
print(f"[Database] Error in default data initialization: {str(e)}")
else:
print("[Database] No DATABASE_URL found, database mode DISABLED (using memory)")

def init_connection_pool(self):
    """Initialize connection pool with proper error handling"""
    if not self.database_url:
        return False

    try:
        print("[Database] Attempting to create connection pool...")
        self.connection_pool = pool.SimpleConnectionPool(
            1, # Min connections
            2, # Max connections (limited to save memory on Render free tier)
            self.database_url,
            connect_timeout=10
        )

        # Test connection to verify it works
        conn = self.connection_pool.getconn()
        if conn:
            print("[Database] Connection test successful")
            self.connection_pool.putconn(conn)
            print("[Database] Connection pool initialized (1-2 connections)")
            return True
        else:
            print("[Database] Connection test failed - no connection returned from pool")
            return False
    except Exception as e:
        print(f"[Database] Connection pool error: {str(e)}")
        traceback.print_exc()
        return False

def get_connection(self):
    """Get connection from pool with retry logic"""
    if not self.connection_pool:
        return None

    try:
        return self.connection_pool.getconn()
    except Exception as e:
        print(f"[Database] Error getting connection: {str(e)}")

    # Try to reinitialize the pool if this fails
    try:
        print("[Database] Attempting to reinitialize connection pool...")
        self.connection_pool = pool.SimpleConnectionPool(1, 2, self.database_url,
            connect_timeout=10)
        return self.connection_pool.getconn()
    except Exception as e:
        print(f"[Database] Failed to reinitialize connection pool: {str(e)}")
        return None

def return_connection(self, conn):
    """Return connection to pool safely"""
    if self.connection_pool and conn:
        try:
            self.connection_pool.putconn(conn)
        except Exception as e:
            print(f"[Database] Error returning connection: {str(e)}")

def init_tables(self):
    """Initialize database tables with indexes for performance"""
    conn = self.get_connection()
    if not conn:
        print("[Database] Failed to get connection for table initialization")
        return

    try:
        cursor = conn.cursor()

        # CVE table
        cursor.execute("""
        CREATE TABLE IF NOT EXISTS cves (
            id VARCHAR(50) PRIMARY KEY,
            description TEXT,

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severity VARCHAR(20),
cvss_score FLOAT,
cwe VARCHAR(50),
published VARCHAR(50),
last_modified VARCHAR(50),
reference_links JSONB,
products JSONB,
metrics JSONB,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
)
"""
self.cached_database_tables['cves'] = True

# Create performance indexes
cursor.execute("CREATE INDEX IF NOT EXISTS idx_cves_severity ON cves(severity)")
cursor.execute("CREATE INDEX IF NOT EXISTS idx_cves_published ON cves(published)")
cursor.execute("CREATE INDEX IF NOT EXISTS idx_cves_cwe ON cves(cwe)")

# Cache metadata table for caching
cursor.execute("""
CREATE TABLE IF NOT EXISTS cache_metadata (
key VARCHAR(100) PRIMARY KEY,
last_updated TIMESTAMP,
total_records INTEGER,
metadata JSONB
)
""")
self.cached_database_tables['cache_metadata'] = True

# Timeline data for chart caching
cursor.execute("""
CREATE TABLE IF NOT EXISTS timeline_data (
year INTEGER,
month INTEGER,
count INTEGER,
PRIMARY KEY (year, month)
)
""")
self.cached_database_tables['timeline_data'] = True

# Summary stats table for chart caching
cursor.execute("""
CREATE TABLE IF NOT EXISTS summary_stats (
key VARCHAR(100) PRIMARY KEY,
count INTEGER,
last_updated TIMESTAMP,
data JSONB
)
""")
self.cached_database_tables['summary_stats'] = True

conn.commit()
print("[Database] Tables initialized successfully")
except Exception as e:
print(f"[Database] Error initializing tables: {str(e)}")
conn.rollback()
finally:
cursor.close()
self.return_connection(conn)

def init_default_data(self):
"""Initialize default data for visualizations if none exists"""
conn = self.get_connection()
if not conn:
print("[Database] Failed to get connection for data initialization")
return

try:
cursor = conn.cursor()

# Check if we have CWE severity data
cursor.execute("SELECT COUNT(*) FROM summary_stats WHERE key = 'cwe_severity'")
cwe_data_exists = cursor.fetchone()[0] > 0

# Check if we have timeline data
cursor.execute("SELECT COUNT(*) FROM timeline_data")

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timeline_data_exists = cursor.fetchone()[0] > 0

# Add CWE severity data if none exists
if not cwe_data_exists:
    print("[Database] Adding default CWE severity data")

# Convert severity data to the format needed for charts
severity_counts = {
    "CRITICAL": 0,
    "HIGH": 0,
    "MEDIUM": 0,
    "LOW": 0
}

for cwe_id, data in DEFAULT_CWE_SEVERITY_DATA.items():
    severity = data["severity"]
    count = data["count"]
    severity_counts[severity] += count

cwe_chart_data = {
    "labels": list(DEFAULT_CWE_SEVERITY_DATA.keys()),
    "datasets": [
        {
            "severity": "CRITICAL",
            "count": severity_counts["CRITICAL"],
            "color": "#ff4d4d"
        },
        {
            "severity": "HIGH",
            "count": severity_counts["HIGH"],
            "color": "#ffaa33"
        },
        {
            "severity": "MEDIUM",
            "count": severity_counts["MEDIUM"],
            "color": "#5599ff"
        },
        {
            "severity": "LOW",
            "count": severity_counts["LOW"],
            "color": "#44cc88"
        }
    ],
    "cwe_data": DEFAULT_CWE_SEVERITY_DATA
}

cursor.execute("""
INSERT INTO summary_stats (key, count, last_updated, data)
VALUES (%s, %s, %s, %s)
ON CONFLICT (key) DO UPDATE SET
count = EXCLUDED.count,
last_updated = EXCLUDED.last_updated,
data = EXCLUDED.data
""", (
    'cwe_severity',
    sum(severity_counts.values()),
    datetime.now(),
    json.dumps(cwe_chart_data)
))

conn.commit()
print("[Database] Default CWE severity data added")

# Add timeline data if none exists
if not timeline_data_exists:
    print("[Database] Adding default timeline data")
    timeline_values = []

for (year, month), count in DEFAULT_TIMELINE_DATA.items():
    timeline_values.append((year, month, count))

# Use batching for better performance
execute_batch(cursor, """
INSERT INTO timeline_data (year, month, count)
VALUES (%s, %s, %s)
ON CONFLICT (year, month) DO UPDATE SET
count = EXCLUDED.count

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"""", timeline_values, page_size=50)

conn.commit()
print(f"[Database] Added timeline data for {len(DEFAULT_TIMELINE_DATA)} months")
except Exception as e:
    print(f"[Database] Error initializing default data: {str(e)}")
    conn.rollback()
finally:
    cursor.close()
self.return_connection(conn)

def save_cves_batch(self, cves: List[Dict[str, Any]]) -> bool:
    """Save multiple CVEs to database with batching for performance"""
    if not self.use_database or not cves:
        return False

    conn = self.get_connection()
    if not conn:
        print("[Database] Failed to get connection for save_cves_batch")
        return False

    try:
        cursor = conn.cursor()

        # Process in batches to manage memory
        batch_size = 50 # Reduced from 100 to conserve memory
        total_saved = 0

        for i in range(0, len(cves), batch_size):
            batch_values = []
            batch = cves[i:i+batch_size]

            for cve in batch:
                batch_values.append((
                    cve.get('ID', ''),
                    cve.get('Description', ''),
                    cve.get('Severity', 'UNKNOWN'),
                    cve.get('CVSS_Score'),
                    cve.get('CWE'),
                    cve.get('Published', ''),
                    cve.get('lastModified', ''),
                    json.dumps(cve.get('References', [])),
                    json.dumps(cve.get('Products', [])),
                    json.dumps(cve.get('metrics', {}))
                ))

            # Use ON CONFLICT to handle duplicates
            execute_batch(cursor, """
INSERT INTO cves (id, description, severity, cvss_score, cwe,
published, last_modified, reference_links, products, metrics)
VALUES (%s, %s, %s, %s, %s, %s, %s, %s, %s, %s)
ON CONFLICT (id) DO UPDATE SET
description = EXCLUDED.description,
severity = EXCLUDED.severity,
cvss_score = EXCLUDED.cvss_score,
cwe = EXCLUDED.cwe,
published = EXCLUDED.published,
last_modified = EXCLUDED.last_modified,
reference_links = EXCLUDED.reference_links,
products = EXCLUDED.products,
metrics = EXCLUDED.metrics,
updated_at = CURRENT_TIMESTAMP
""", batch_values)

            conn.commit()
            total_saved += len(batch)

        # Clear batch to free memory
        batch_values = []
        gc.collect()

        print(f"[Database] Saved {total_saved} CVEs to database")
        return True
    except Exception as e:
        print(f"[Database] Error saving CVEs: {str(e)}")
        conn.rollback()
        return False

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finally:
    cursor.close()
    self.return_connection(conn)

def get_cves_by_filter(self, year=None, month=None, day=None, severity=None,
    limit=1000):
    """Get CVEs with filters using indexes for performance"""
    if not self.use_database:
        # Return default data if database is not available
        return self._get_default_cves(year, month, day, severity, limit)

    conn = self.get_connection()
    if not conn:
        print("[Database] Failed to get connection for get_cves_by_filter")
        # Fall back to default data
        return self._get_default_cves(year, month, day, severity, limit)

    try:
        cursor = conn.cursor(cursor_factory=RealDictCursor)

        # Build query based on filters
        query = """
        SELECT id, description, severity, cvss_score, cwe,
        published, last_modified, reference_links, products, metrics
        FROM cves
        WHERE 1=1
        """
        params = []

        if year:
            query += " AND published LIKE %s"
            params.append(f"{year}%")

        if month:
            query += " AND published LIKE %s"
            month_str = f"{int(month):02d}"
            params.append(f"{year}-{month_str}%")

        if day:
            query += " AND published LIKE %s"
            day_str = f"{int(day):02d}"
            params.append(f"{year}-{month_str}-{day_str}%")

        if severity:
            query += " AND severity = %s"
            params.append(severity.upper())

        # Reduce limit for memory optimization
        limit = min(limit, 1000)
        query += " ORDER BY published DESC LIMIT %s"
        params.append(limit)

        cursor.execute(query, params)
        rows = cursor.fetchall()

        # Convert to list of dicts
        cves = []
        for row in rows:
            try:
                # Handle JSON fields safely
                reference_links = row['reference_links']
                if isinstance(reference_links, str):
                    reference_links = json.loads(reference_links)

                products = row['products']
                if isinstance(products, str):
                    products = json.loads(products)

                metrics = row['metrics']
                if isinstance(metrics, str):
                    metrics = json.loads(metrics)

                cve = {
                    'ID': row['id'],
                    'Description': row['description'],
                    'Severity': row['severity'],
                    'CVSS_Score': row['cvss_score'],

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'CWE': row['cwe'],
'Published': row['published'],
'lastModified': row['last_modified'],
'References': reference_links if isinstance(reference_links, list) else [],
'Products': products if isinstance(products, list) else [],
'metrics': metrics if isinstance(metrics, dict) else {}
}
cves.append(cve)
except Exception as e:
print(f"[Database] Error processing row: {str(e)}")

return cves
except Exception as e:
print(f"[Database] Error fetching CVEs with filter: {str(e)}")
# Fall back to default data
return self._get_default_cves(year, month, day, severity, limit)
finally:
cursor.close()
self.return_connection(conn)

def _get_default_cves(self, year=None, month=None, day=None, severity=None, limit=1000):
"""Provide default CVEs when database fails"""
# Generate some placeholder data based on the year/month/day
current_year = datetime.now().year
cves = []

# If no year specified, use current year
if not year:
year = current_year

# Generate severities based on DEFAULT_CWE_SEVERITY_DATA distribution
severity_counts = {"CRITICAL": 0, "HIGH": 0, "MEDIUM": 0, "LOW": 0, "UNKNOWN": 0}
for _, data in DEFAULT_CWE_SEVERITY_DATA.items():
severity_counts[data["severity"]] += 1

total_count = sum(severity_counts.values())
severity_probs = {k: v / total_count for k, v in severity_counts.items()}

# Generate about 1000 CVEs for current year (reduced from 4000), 500 for previous year
count = 1000 if int(year) == current_year else 500
count = min(count, limit)

# Distribution across months (more recent months have more CVEs)
month_weights = {
1: 0.7, 2: 0.8, 3: 0.8, 4: 0.9, 5: 0.9,
6: 1.0, 7: 1.0, 8: 1.1, 9: 1.1, 10: 1.2, 11: 1.2, 12: 1.3
}

# Generate mock CVEs
for i in range(count):
# Determine severity using weighted probability
rand = (i % 100) / 100
cumulative = 0
cve_severity = "UNKNOWN"

for sev, prob in severity_probs.items():
cumulative += prob
if rand <= cumulative:
cve_severity = sev
break

# Determine month based on weights (if no month specified)
if not month:
rand_month = 1 + (i % 12)
else:
rand_month = int(month)

# Determine day (if no day specified)
if not day:
rand_day = 1 + (i % 28)
else:
rand_day = int(day)

# Skip if this doesn't match filter criteria
if severity and severity.upper() != cve_severity:
continue

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# Create mock CVE
cve_id = f"CVE-{year}-{1000 + i}"

cves.append({
    'ID': cve_id,
    'Description': f"Mock vulnerability for testing purposes ({cve_id})",
    'Severity': cve_severity,
    'CVSS_Score': 7.5 if cve_severity == "HIGH" else
    9.1 if cve_severity == "CRITICAL" else
    5.5 if cve_severity == "MEDIUM" else 3.2,
    'CWE': list(DEFAULT_CWE_SEVERITY_DATA.keys())[i % len(DEFAULT_CWE_SEVERITY_DATA)],
    'Published': f"{year}-{rand_month:02d}-{rand_day:02d}",
    'lastModified': f"{year}-{rand_month:02d}-{rand_day:02d}",
    'References': [],
    'Products': [],
    'metrics': {}
})

return cves

def get_timeline_data(self, years=1) -> Dict[str, Any]:
    """Get timeline data from database with fallback to defaults"""
    if not self.use_database:
        return self.get_default_timeline_data(years)

    conn = self.get_connection()
    if not conn:
        print("[Database] Failed to get connection for get_timeline_data")
        return self.get_default_timeline_data(years)

    try:
        cursor = conn.cursor()

        # Get current year
        current_year = datetime.now().year

        # Calculate start year
        start_year = current_year - years

        cursor.execute("""
        SELECT year, month, count
        FROM timeline_data
        WHERE year >= %s
        ORDER BY year, month
        """, (start_year,))

        rows = cursor.fetchall()

        # Process into timeline format
        labels = []
        values = []
        raw_data = {}
        total_cves = 0

        for year, month, count in rows:
            label = f"{year}-{month:02d}"
            labels.append(label)
            values.append(count)
            raw_data[label] = count
            total_cves += count

        # If no data was found, use default data
        if not rows:
            print("[Database] No timeline data found, using defaults")
            return self.get_default_timeline_data(years)

        return {
            'labels': labels,
            'values': values,
            'total_cves': total_cves,
            'months_covered': len(labels),
            'raw_data': raw_data
        }
    except Exception as e:
        print(f"[Database] Error fetching timeline data: {str(e)}")
        return self.get_default_timeline_data(years)
    finally:

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cursor.close()
self.return_connection(conn)

def get_default_timeline_data(self, years=1) -> Dict[str, Any]:
    """Get default timeline data for visualization when database fails"""
    labels = []
    values = []
    raw_data = {}
    total_cves = 0

    # Use default timeline data to ensure chart works
    current_year = datetime.now().year
    start_year = current_year - years

    for (year, month), count in DEFAULT_TIMELINE_DATA.items():
        if year >= start_year:
            label = f"{year}-{month:02d}"
            labels.append(label)
            values.append(count)
            raw_data[label] = count
            total_cves += count

    return {
        'labels': labels,
        'values': values,
        'total_cves': total_cves,
        'months_covered': len(labels),
        'raw_data': raw_data
    }

def save_summary_stats(self, key, count, data):
    """Save summary statistics to database with TTL for caching"""
    if not self.use_database:
        return False

    conn = self.get_connection()
    if not conn:
        print(f"[Database] Failed to get connection for save_summary_stats: {key}")
        return False

    try:
        cursor = conn.cursor()

        cursor.execute("""
        INSERT INTO summary_stats (key, count, last_updated, data)
        VALUES (%s, %s, CURRENT_TIMESTAMP, %s)
        ON CONFLICT (key) DO UPDATE SET
        count = EXCLUDED.count,
        last_updated = CURRENT_TIMESTAMP,
        data = EXCLUDED.data
        """, (key, count, json.dumps(data)))

        conn.commit()
        return True
    except Exception as e:
        print(f"[Database] Error saving summary stats: {str(e)}")
        conn.rollback()
        return False
    finally:
        cursor.close()
        self.return_connection(conn)

def get_summary_stats(self, key):
    """Get summary statistics from database with TTL check"""
    if not self.use_database:
        # Return default data based on key
        if key == 'cwe_severity':
            return self.get_default_cwe_severity_stats()
        return None

    conn = self.get_connection()
    if not conn:
        print(f"[Database] Failed to get connection for get_summary_stats: {key}")
        if key == 'cwe_severity':
            return self.get_default_cwe_severity_stats()
        return None

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try:
    cursor = conn.cursor(cursor_factory=RealDictCursor)

    cursor.execute("""
    SELECT count, last_updated, data
    FROM summary_stats
    WHERE key = %s
    """, (key,))

    row = cursor.fetchone()
    if row:
        # Check TTL (24 hours instead of 6 to reduce database load)
        last_updated = row['last_updated']
        now = datetime.now()

        if isinstance(last_updated, str):
            # Convert string to datetime
            last_updated = datetime.fromisoformat(last_updated.replace('Z', '+00:00'))

        age = now - last_updated

        if age.total_seconds() < 24 * 60 * 60: # 24 hours
            return {
                'count': row['count'],
                'last_updated': row['last_updated'],
                'data': row['data']
            }
        else:
            print(f"[Database] Cache for {key} expired ({age.total_seconds()/3600:.1f} hours old)")

    # If no data found or cache expired, return defaults based on key
    if key == 'cwe_severity':
        return self.get_default_cwe_severity_stats()
    return None
except Exception as e:
    print(f"[Database] Error getting summary stats: {str(e)}")

# Return default data based on key
if key == 'cwe_severity':
    return self.get_default_cwe_severity_stats()
return None
finally:
    cursor.close()
self.return_connection(conn)

def get_default_cwe_severity_stats(self):
    """Get default CWE severity stats for visualization when database fails"""
    # Calculate severity counts from default data
    severity_counts = {
        "CRITICAL": 0,
        "HIGH": 0,
        "MEDIUM": 0,
        "LOW": 0
    }

    for cwe_id, data in DEFAULT_CWE_SEVERITY_DATA.items():
        severity = data["severity"]
        count = data["count"]
        severity_counts[severity] += count

    # Format data for chart
    cwe_chart_data = {
        "labels": list(DEFAULT_CWE_SEVERITY_DATA.keys()),
        "datasets": [
            {
                "severity": "CRITICAL",
                "count": severity_counts["CRITICAL"],
                "color": "#ff4d4d"
            },
            {
                "severity": "HIGH",
                "count": severity_counts["HIGH"],
                "color": "#ffaa33"
            },
            {
                "severity": "MEDIUM",
                "count": severity_counts["MEDIUM"],

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"color": "#5599ff"
},
{
"severity": "LOW",
"count": severity_counts["LOW"],
"color": "#44cc88"
}
},
"cwe_data": DEFAULT_CWE_SEVERITY_DATA
}

return {
'count': sum(severity_counts.values()),
'last_updated': datetime.now().isoformat(),
'data': cwe_chart_data
}

def get_stats(self) -> Dict[str, Any]:
"""Get database statistics for monitoring"""
if not self.use_database:
return {'database_enabled': False, 'memory_mode': True}

conn = self.get_connection()
if not conn:
print("[Database] Failed to get connection for get_stats")
return {'database_enabled': False, 'connection_failed': True}

try:
cursor = conn.cursor()

cursor.execute("SELECT COUNT(*) FROM cves")
total_cves = cursor.fetchone()[0]

# Get table sizes (if possible)
try:
cursor.execute("""
SELECT relname as table, pg_size_pretty(pg_total_relation_size(relid)) as size
FROM pg_catalog.pg_statio_user_tables
ORDER BY pg_total_relation_size(relid) DESC
""")
tables = {row[0]: row[1] for row in cursor.fetchall()}
except:
tables = {}

# Check if default data is loaded
cursor.execute("SELECT COUNT(*) FROM timeline_data")
timeline_count = cursor.fetchone()[0]

cursor.execute("SELECT COUNT(*) FROM summary_stats WHERE key = 'cwe_severity'")
cwe_data_exists = cursor.fetchone()[0] > 0

return {
'total_cves': total_cves,
'database_enabled': True,
'tables': list(tables.keys()),
'table_sizes': tables,
'timeline_data_loaded': timeline_count > 0,
'cwe_data_loaded': cwe_data_exists
}
except Exception as e:
print(f"[Database] Error getting stats: {str(e)}")
return {'database_enabled': False, 'error': str(e)}
finally:
cursor.close()
self.return_connection(conn)

def close(self):
"""Close all connections in pool"""
if self.connection_pool:
self.connection_pool.closeall()
print("[Database] Connection pool closed")

# Adding the missing methods that caused the errors
def get_cache_metadata(self, cache_key):
"""Get metadata for cached data - THIS WAS MISSING"""
if not self.use_database:
return None

```

```

conn = self.get_connection()
if not conn:
print(f"[Database] Failed to get connection for get_cache_metadata: {cache_key}")
return None

try:
cursor = conn.cursor(cursor_factory=RealDictCursor)

cursor.execute("""
SELECT key, last_updated, total_records, metadata
FROM cache_metadata
WHERE key = %s
""", (cache_key,))

row = cursor.fetchone()
return row
except Exception as e:
print(f"[Database] Error getting cache metadata: {str(e)}")
return None
finally:
cursor.close()
self.return_connection(conn)

def update_cache_metadata(self, cache_key, total_records, metadata=None):
"""Update cache metadata - THIS WAS MISSING"""
if not self.use_database:
return False

conn = self.get_connection()
if not conn:
print(f"[Database] Failed to get connection for update_cache_metadata: {cache_key}")
return False

try:
cursor = conn.cursor()

cursor.execute("""
INSERT INTO cache_metadata (key, last_updated, total_records, metadata)
VALUES (%s, CURRENT_TIMESTAMP, %s, %s)
ON CONFLICT (key) DO UPDATE SET
last_updated = CURRENT_TIMESTAMP,
total_records = EXCLUDED.total_records,
metadata = EXCLUDED.metadata
""", (cache_key, total_records, json.dumps(metadata or {})))

conn.commit()
return True
except Exception as e:
print(f"[Database] Error updating cache metadata: {str(e)}")
conn.rollback()
return False
finally:
cursor.close()
self.return_connection(conn)

def get_recent_cves(self, days=30, limit=25):
"""Get recent CVEs for the learn pages - THIS WAS MISSING"""
if not self.use_database:
return self._get_default_cves(None, None, None, None, limit)

conn = self.get_connection()
if not conn:
print("[Database] Failed to get connection for get_recent_cves")
return self._get_default_cves(None, None, None, None, limit)

try:
cursor = conn.cursor(cursor_factory=RealDictCursor)

# Calculate the date threshold (last X days)
from datetime import datetime, timedelta
threshold_date = (datetime.now() - timedelta(days=days)).strftime('%Y-%m')

query = """
SELECT id, description, severity, cvss_score, cwe, published
FROM cves
WHERE published >= %s

```

```
ORDER BY published DESC
LIMIT %s
"""
```

```
cursor.execute(query, (threshold_date, limit))
rows = cursor.fetchall()
```

```
# Convert to list of dicts in the expected format
```

```
cves = []
for row in rows:
    cve = {
        'ID': row['id'],
        'Description': row['description'],
        'Severity': row['severity'],
        'CVSS_Score': row['cvss_score'],
        'CWE': row['cwe'],
        'Published': row['published']
    }
    cves.append(cve)
```

```
return cves
except Exception as e:
    print(f"[Database] Error getting recent CVEs: {str(e)}")
    return self._get_default_cves(None, None, None, None, limit)
finally:
    cursor.close()
    self.return_connection(conn)
```

```
def get_cve_detail(self, cve_id):
    """Get details for a specific CVE"""
    if not self.use_database:
        return None
```

```
conn = self.get_connection()
if not conn:
    print(f"[Database] Failed to get connection for get_cve_detail: {cve_id}")
    return None
```

```
try:
    cursor = conn.cursor(cursor_factory=RealDictCursor)
```

```
cursor.execute("""
SELECT id, description, severity, cvss_score, cwe,
published, last_modified, reference_links, products, metrics
FROM cves
WHERE id = %s
""", (cve_id,))
```

```
row = cursor.fetchone()
if not row:
    return None
```

```
# Process JSON fields
reference_links = row['reference_links']
if isinstance(reference_links, str):
    reference_links = json.loads(reference_links)
```

```
products = row['products']
if isinstance(products, str):
    products = json.loads(products)
```

```
metrics = row['metrics']
if isinstance(metrics, str):
    metrics = json.loads(metrics)
```

```
return {
    'ID': row['id'],
    'Description': row['description'],
    'Severity': row['severity'],
    'CVSS_Score': row['cvss_score'],
    'CWE': row['cwe'],
    'Published': row['published'],
    'lastModified': row['last_modified'],
    'References': reference_links if isinstance(reference_links, list) else [],
    'Products': products if isinstance(products, list) else [],
    'metrics': metrics if isinstance(metrics, dict) else {}
}
```

```

except Exception as e:
    print(f"[Database] Error getting CVE detail: {str(e)}")
    return None
finally:
    cursor.close()
    self.return_connection(conn)

def save_cve_detail(self, cve):
    """Save a single CVE detail"""
    if not self.use_database or not cve:
        return False

    return self.save_cves_batch([cve])

def get_all_cves(self, limit=1000):
    """Get all CVEs with a limit"""
    return self.get_cves_by_filter(limit=limit)

def clear_all_cves(self):
    """Clear all CVEs from the database (for cache clearing)"""
    if not self.use_database:
        return False

    conn = self.get_connection()
    if not conn:
        print("[Database] Failed to get connection for clear_all_cves")
        return False

    try:
        cursor = conn.cursor()

        cursor.execute("DELETE FROM cves WHERE id NOT LIKE 'RESERVED%'")
        conn.commit()

        cursor.execute("DELETE FROM cache_metadata")
        conn.commit()

        print("[Database] Cleared all CVEs and cache metadata")
        return True
    except Exception as e:
        print(f"[Database] Error clearing CVEs: {str(e)}")
        conn.rollback()
        return False
    finally:
        cursor.close()
        self.return_connection(conn)

def save_timeline_data(self, year_month_counts):
    """Save timeline data to database"""
    if not self.use_database or not year_month_counts:
        return False

    conn = self.get_connection()
    if not conn:
        print("[Database] Failed to get connection for save_timeline_data")
        return False

    try:
        cursor = conn.cursor()

        # Prepare values for batch insert
        values = []
        for (year, month), count in year_month_counts.items():
            values.append((year, month, count))

        # Use batching for better performance
        execute_batch(cursor, """
INSERT INTO timeline_data (year, month, count)
VALUES (%s, %s, %s)
ON CONFLICT (year, month) DO UPDATE SET
count = EXCLUDED.count
""", values, page_size=50)

        conn.commit()
        print(f"[Database] Saved timeline data for {len(values)} months")
        return True
    except Exception as e:

```

```
print(f"[Database] Error saving timeline data: {str(e)}")
conn.rollback()
return False
finally:
    cursor.close()
    self.return_connection(conn)

# Global database manager instance
db_manager = DatabaseManager()
```